

— A topological graph:

graph drawn in the plane where vertices are represented by distinct points and edges are Jordan arcs connecting the points

$$f: [0, 1] \rightarrow \mathbb{R}^2$$

continuous
and injective

$$\forall x, y \in [0, 1] \quad x \neq y \Rightarrow f(x) \neq f(y)$$

not passing through any other vertex,
any pair edges crosses a finite number
of times

— Embedding of a graph: edge crossing-free drawing

— Weak H-T for Planar Drawings —

TH1 : h has
an x -monotone
and even
drawing

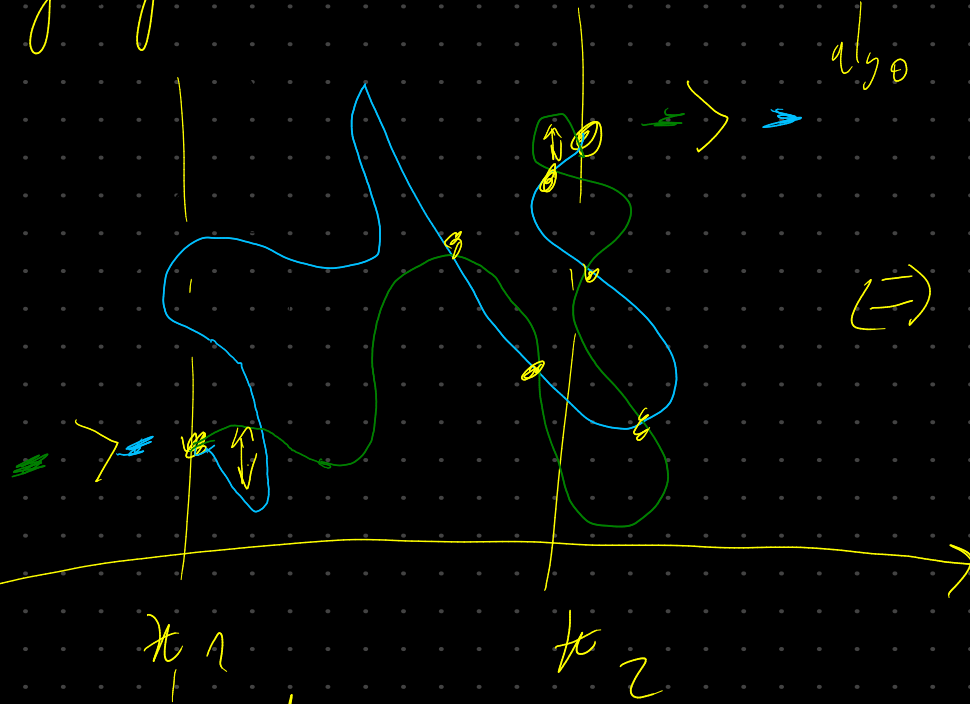
$\Rightarrow$

h has x -monotone
embedding

with notation system
coordinates

stay the same

proof of TH 1



even crossings

same order they enter x_1 and x_2



$h \in \{ \text{inf } H_f \}$
has also
 x -monotone
embedding
with same
 π
 $\downarrow$
relative x -order

- Proof:



i) h consists of multiple components

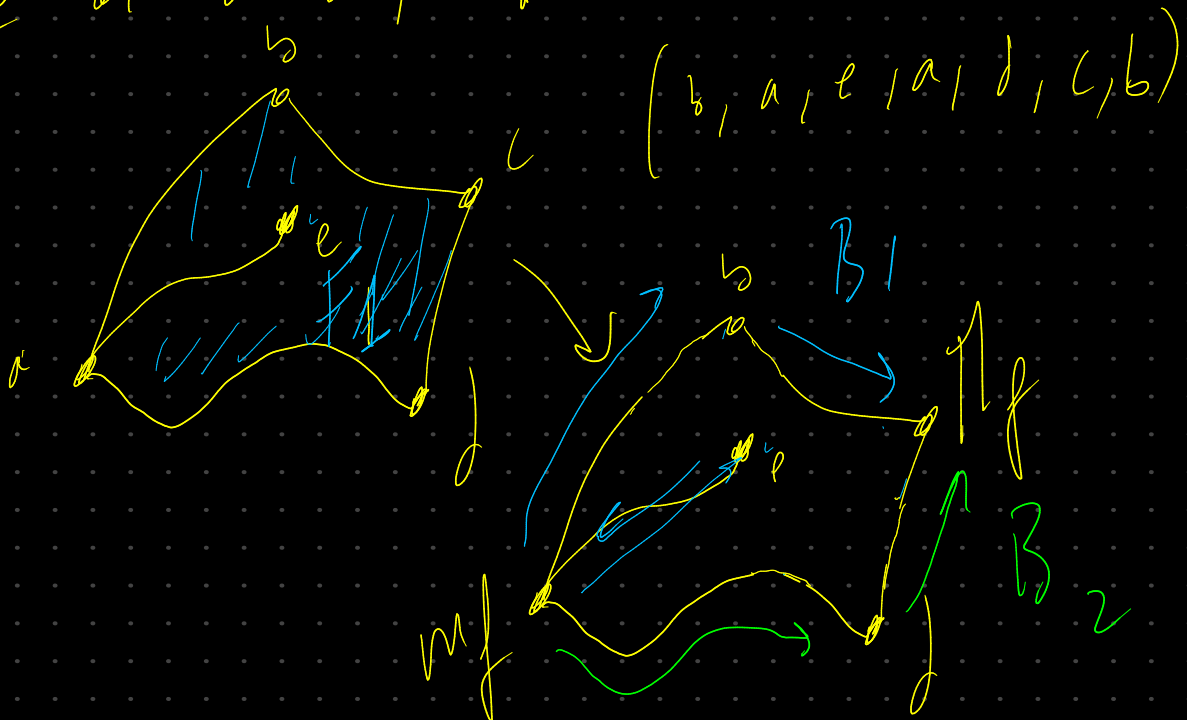


we can't
shift vertically

(for now let's assume that's true)

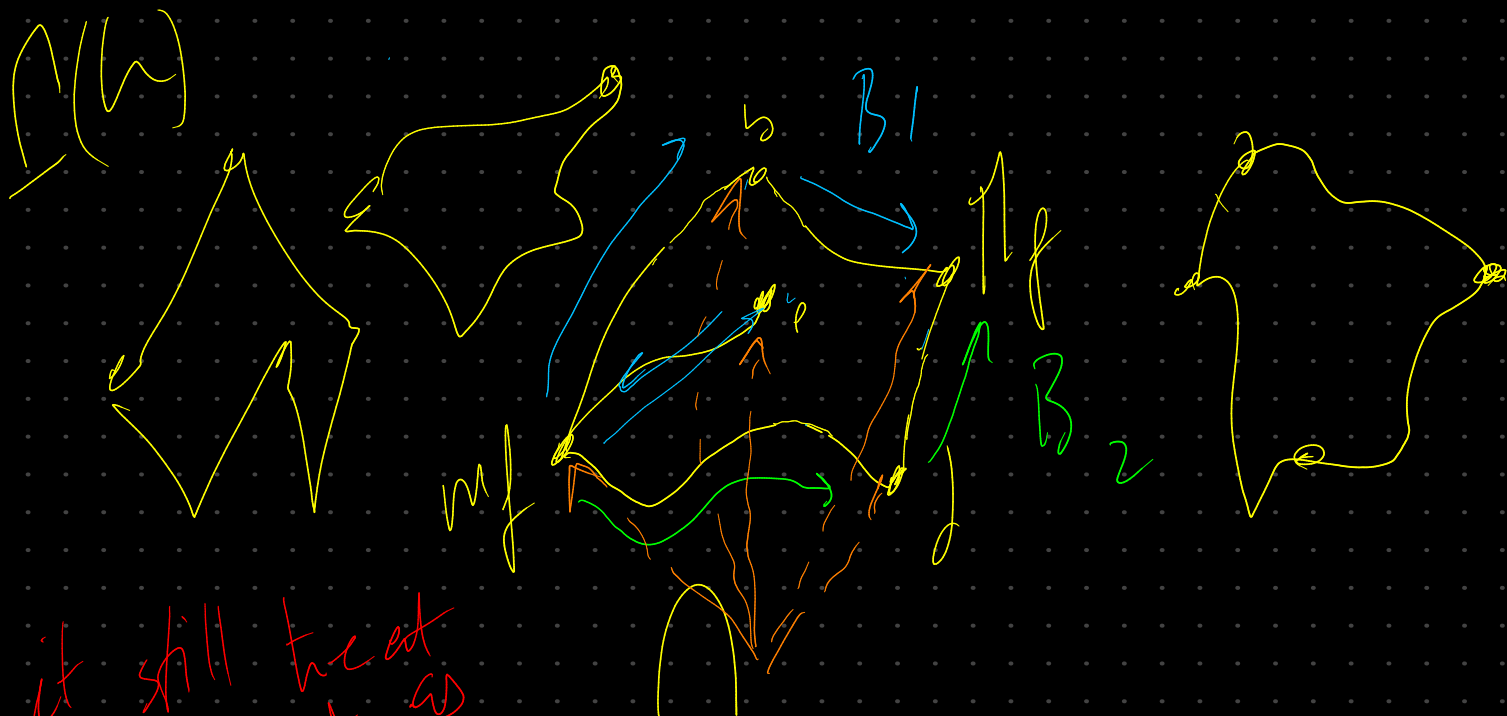
we can treat each connected component as
a graph by itself and it's connected.

The change in the embeddings won't affect
the other components.

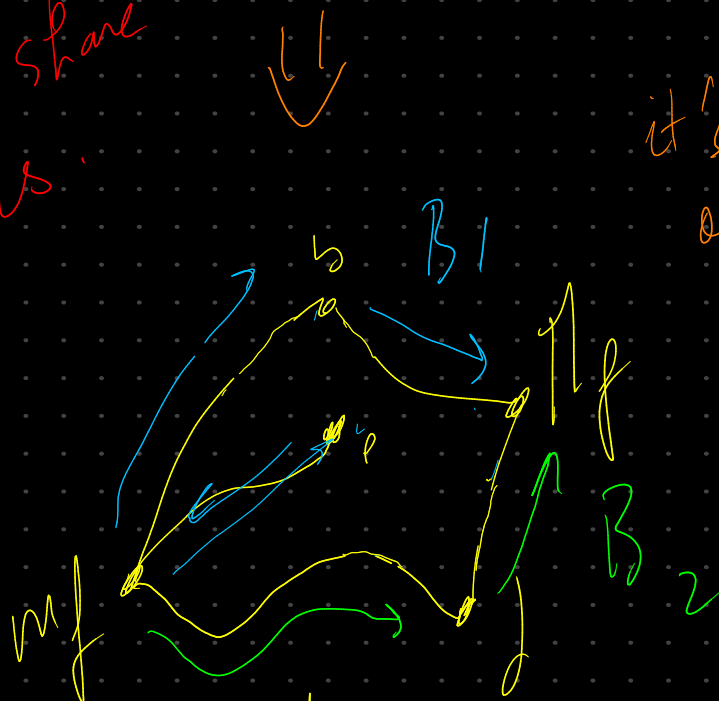


let D_f be drawing of h interested with

$$\Delta_f = \{ (x, y) \in \mathbb{R}^2 : \kappa(m_f) < x < \kappa(n_f) \}$$



it still treat
C components as
if they don't share
some levels.



it's a subset
of a plane
not a
graph

Next step: locally redraw h in Δ_f so
 $m_f n_f$ can be inserted as a straight line seg.
why?

$\forall$ (topologically) connected component Z of D_f

$\forall \kappa$

$$\kappa(m_f) < \kappa < \kappa(\Pi_f)$$

$\exists y \in B_1$ at κ

$$y = \min_{y \in Y(Z)} y(\kappa)$$



κ

the lowest
pt of the level
of D_f in κ

$\forall \kappa \in X(Z)$

$$\kappa(m_f) < \kappa < \kappa(\Pi_f)$$

$\exists y' \in B_2$ at κ

$$y' = \max_{y \in Y(Z)} y(\kappa)$$

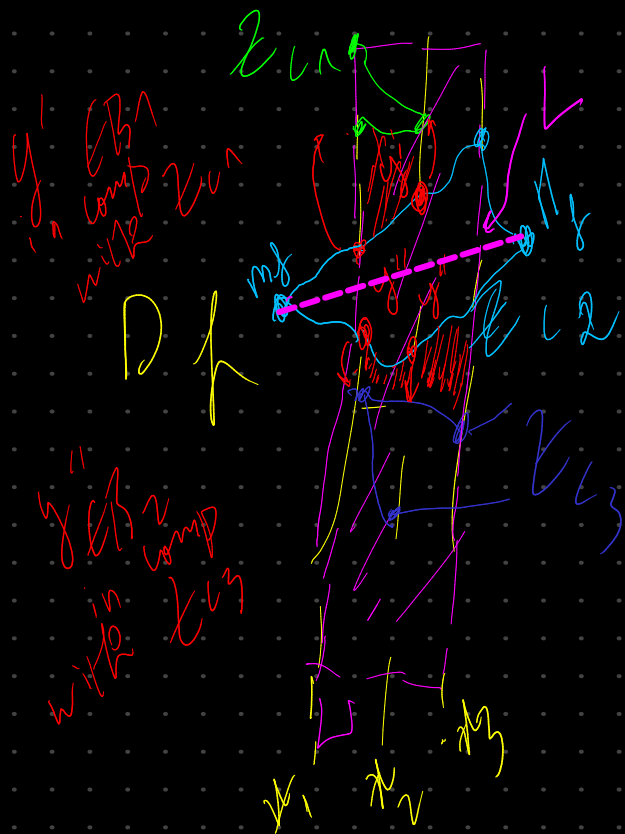


κ

why
either?

let Z_1 be the union of all the components of the first type & Z_2 for the 2nd type.

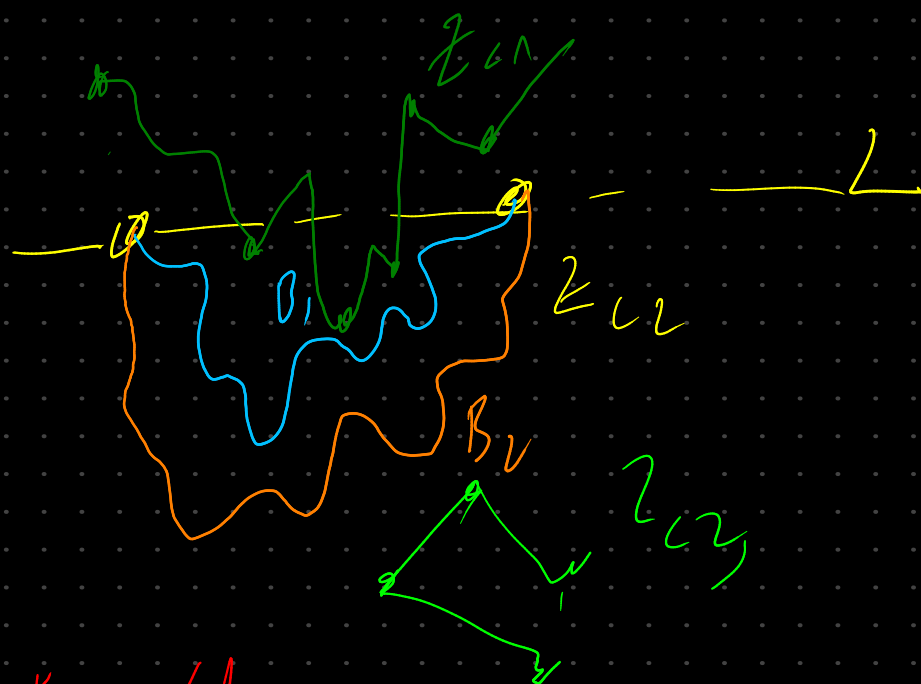
Let L be the line through m_f and M_f


$$x \in \mathcal{X}(M_f), (x(\Pi_f))^\perp$$

$\mathbb{Z}_2^2 \times \mathbb{Z}_2$ at x

if we take $z \in \mathbb{C}$

M_3 example doesn't show the necessity of the part of the proof that we are going to handle.



objective:

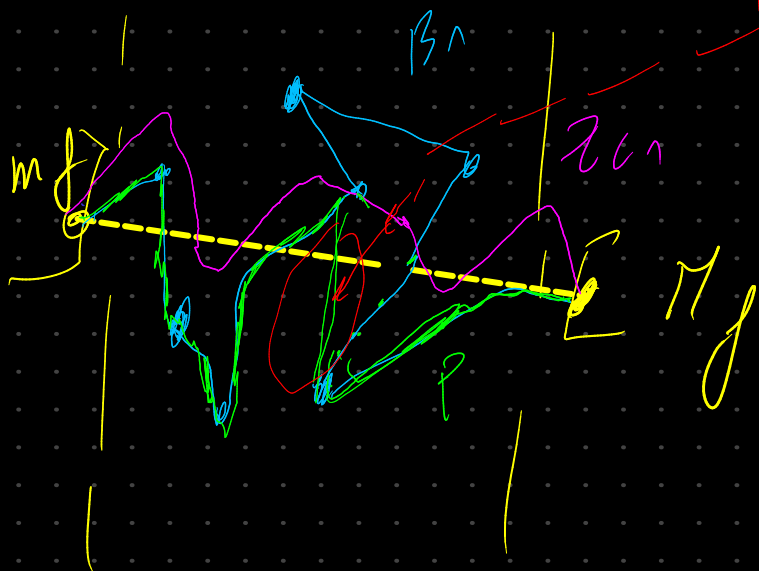
Z_{C1} to half-plane above L .

Z_{C2} to half-plane under L

I am taking an example with 1 component $\rightarrow$ above under

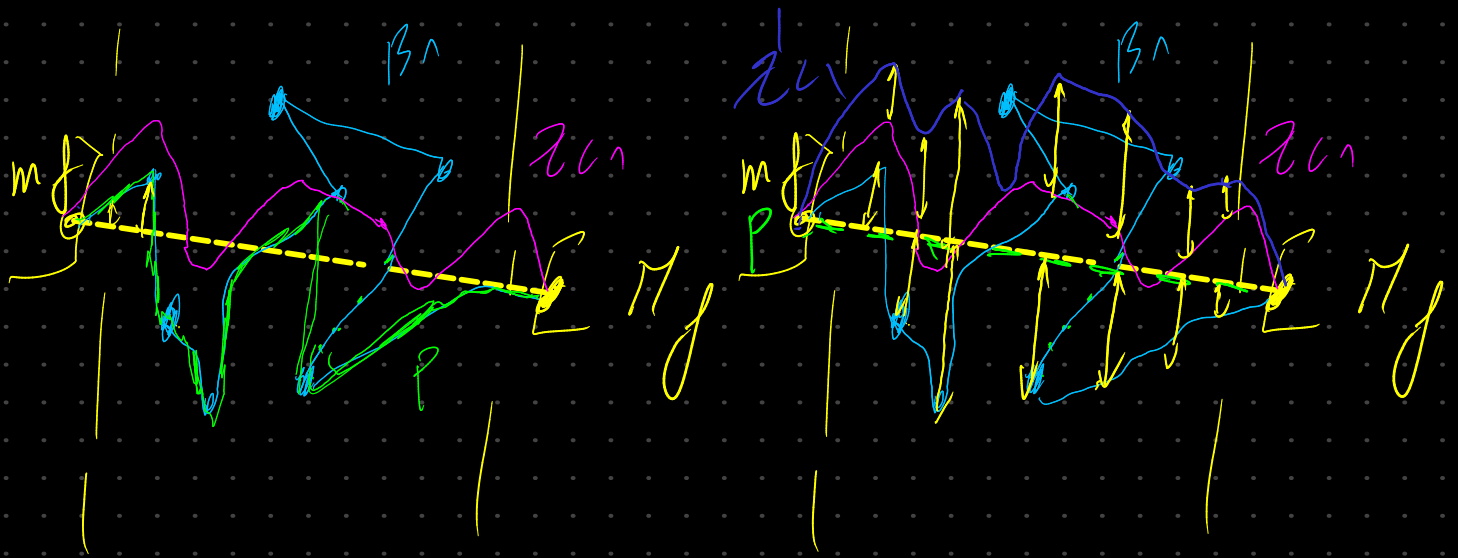
without changing x -value of any point in Z_{C1} Z_{C2}

P an x -monotone curve (m_f, M_f) below Z_{C1} in L_f



how should i represent this part?

move Z_{C1} up by vertical distance between P and L at $x = x(v)$



it's like taking P
flat with L
so we are sure
that z_{cn} is
above L in D_f

z_{cn} is after the c to
with distance $g(P-L)$

but actually at the
same time we are
pushing B_1 up

Goal: let (m, p) edge not cross any
component when added

$\Rightarrow h \cup \{(m, p)\}$ is
also π -monotone embedding.

$G / \{v_1, \dots, v_i\}$ with

the j -th leftmost vertex of G is connected.



we can arrange $(v_1, v_2, v_4, v_3, v_5)$ what
with $G - \{v_1, \dots, v_n\}$ not being connected?

G has an x -monotone & even drawing $\Rightarrow$

G has an x -monotone embedding with
 $\left. \begin{array}{l} \text{ } \end{array} \right\} \begin{array}{l} \text{ } \end{array}$ v keeps x -coordinate
 same rotation system

* something stronger is true?

All even edges can be made cross-free in the drawing of a graph which contains odd edges, while maintaining being monotone.

$(u, v) \in E(G)$ is bounded if

(interior point p) of uv $x(u) < x(p) < x(v)$

x x -monotone

$\forall e \in E(G)$
cross $\forall$ vertical
line at
most once

$\forall$ vertical line
contains at most
one vertex.



e is
bounded
doesn't need
to be x -monotone
inside the x -band

Lemma 2:

G with bounded edge



e can be redrawn

(without)

changing the $\Gamma_u \setminus \Gamma_u(e)$

pos of e in

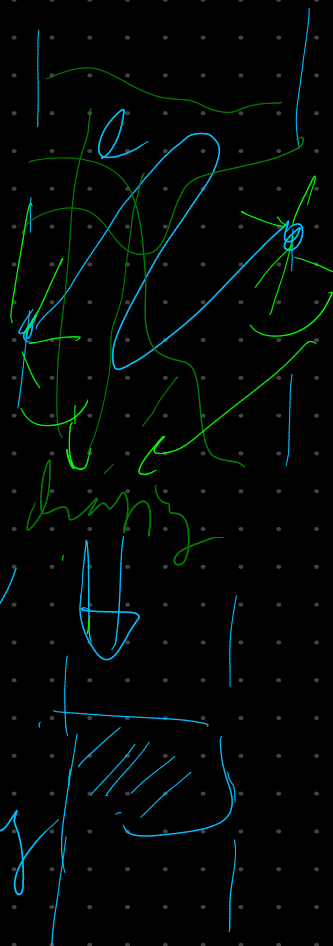
the rotations of the endpoints

(so that)

e is π -monotone

↓ $\forall e' \in E(u) \setminus e$
parity of crossing

between e and e' not changing

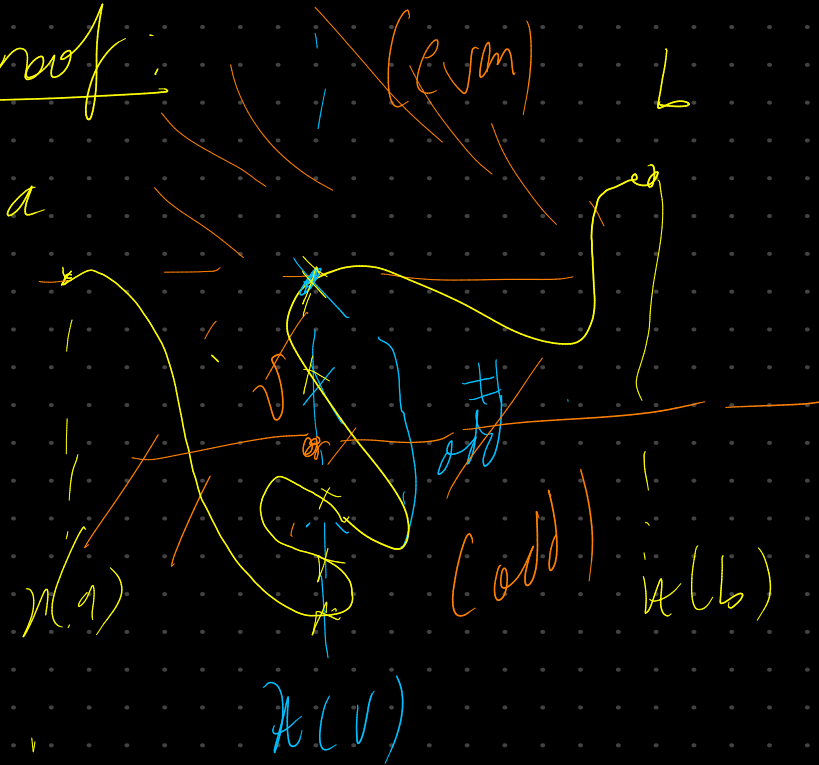


constraints

don't change
 $n \setminus \{e\}$

no change
 $\text{on } (u, v) = e$
 rotations
 in or out v .

proof:



$$v \in V(h)$$

$$h(a) \leq h(v) \leq h(b)$$

v should not
 be on the
 edge (a, b)

$\forall l \in [1, n] \exists! v \in V(h)$, is there
 no crossing of vertices when we talk
 about an edge? i think yes.

